

Computer Science & Programming Practice: 150 Questions

Technical Education Workshop

1 Basics of Computer (30 Questions)

1. Which of the following is known as the "Brain" of the computer?
 - A) RAM
 - B) ROM
 - C) CPU
 - D) Hard Disk
2. What is the full form of ALU?
 - A) Arithmetic Logic Unit
 - B) Allowed Logic Unit
 - C) Array Logic Unit
 - D) Arithmetic Layer Unit
3. Which memory is volatile in nature?
 - A) ROM
 - B) RAM
 - C) PROM
 - D) EPROM
4. 1 Terabyte (TB) is equal to:
 - A) 1024 GB
 - B) 1000 MB
 - C) 1024 KB
 - D) 1024 MB
5. Which of the following is an Operating System?
 - A) MS Word
 - B) Google Chrome
 - C) Windows 11
 - D) Oracle
6. *Note: Questions 6–30 cover input/output devices, networking basics, and computer generations.*

2 C Programming (30 Questions)

7. Who is the father of C language?
- A) Bjarne Stroustrup
 - B) Dennis Ritchie
 - C) James Gosling
 - D) Guido van Rossum
8. Which operator is used for the "Address of" operation?
- A) *
 - B) &
 - C) %
 - D) !
9. What is the correct way to declare a pointer in C?
- A) int ptr;
 - B) int &ptr;
 - C) int *ptr;
 - D) int \$ptr;
10. Which function is used to read a character from the console?
- A) printf()
 - B) scanf()
 - C) getch()
 - D) read()
11. The size of an 'int' data type in C is typically:
- A) 1 Byte
 - B) 2 or 4 Bytes
 - C) 8 Bytes
 - D) Depends on the OS
12. *Note: Questions 16–30 focus on loops, arrays, structures, and file handling in C.*

3 Java Programming (30 Questions)

13. Which of these is NOT a feature of Java?
- A) Platform Independent
 - B) Object Oriented
 - C) Pointers
 - D) Robust
14. What is the extension of a compiled Java class file?

- A) .java
- B) .class
- C) .txt
- D) .obj

15. Which keyword is used to inherit a class in Java?

- A) implements
- B) inherits
- C) extends
- D) super

16. Which of the following is used to handle exceptions in Java?

- A) try-catch
- B) do-while
- C) switch-case
- D) if-else

17. What does JVM stand for?

- A) Java Variable Machine
- B) Java Visual Machine
- C) Java Virtual Machine
- D) Java Verified Machine

18. *Note: Questions 26–30 cover Multithreading, Interfaces, and Collections in Java.*

4 Python Programming (30 Questions)

19. Which of the following is the correct file extension for Python files?

- A) .py
- B) .python
- C) .pt
- D) .p

20. Which keyword is used to create a function in Python?

- A) function
- B) def
- C) fun
- D) define

21. What is the output of `print(2 ** 3)`?

- A) 6
- B) 8
- C) 9

D) 5

22. Which data structure in Python is immutable?

- A) List
- B) Dictionary
- C) Set
- D) Tuple

23. How do you start a comment in Python?

- A) //
- B) /*
- C) #
- D) %

24. *Note: Questions 36–60 involve list comprehensions, slicing, and Python decorators.*

5 Data Structures (30 Questions)

25. Which data structure follows the LIFO (Last In First Out) principle?

- A) Queue
- B) Linked List
- C) Stack
- D) Tree

26. The process of visiting all nodes in a tree is called:

- A) Searching
- B) Traversal
- C) Sorting
- D) Inserting

27. Which of the following is a non-linear data structure?

- A) Array
- B) Linked List
- C) Graph
- D) Stack

28. Time complexity of Binary Search is:

- A) $O(n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(1)$

29. Which sorting algorithm has the best average-case time complexity?

- A) Bubble Sort

- B) Insertion Sort
- C) Quick Sort
- D) Selection Sort

30. *Note: Questions 131–150 cover Hashing, AVL Trees, and B-Trees.*